

Climate 5000 Series - Gen 4

18K BTU Single Zone Max Performance System

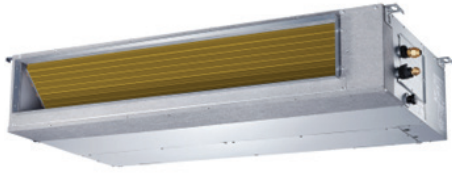
Ducted IDU

Engineering
 Submittal
 Sheet



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Overview and Project Information



Project name	
Location	
Architect	
Engineer	
General contractor	
Reference #	

18K Btu Single Zone Max Performance System - Ducted IDU

Indoor part number		8-733-962-711
Indoor model number		BMS500-AAU018-1AHDXD
Outdoor part number		8-733-962-690
Outdoor model number		BMS500-AAS018-1CSXHD
Efficiency Data		AHRI 210/240 2023
SEER2		18.0
EER2		12.0
HSPF2 Region IV		10.5
HSPF2 Region V		9.0
COP(47F)		3.37
COP (5F)		2.00
Performance Data		AHRI 210/240 2023
Cooling	Capacity Range	6500~18800
	Rated Capacity	17000
	Current	9.1
	Input Power	1420
Heating	Capacity Range	8900~22000
	Rated Capacity (47F)	19000
	Rated Capacity (5F)	18000
	Current	10.5
	Input Power	1650
Electrical Data		
Power supply (V / Ø / Hz)		208-230 / 1 / 60
Min. circuit ampacity indoor/outdoor (A)		3.0 / 19.0
Max. fuse indoor/outdoor (A)		15.0 / 20.0
Compressor Data		
Type:		Rotary
Refrigerant oil /oil charge (ml)		VG74 / 620
Input (W)		805 / 2045
Rated current (RLA/A)		4.15 / 9.30
Piping		
Refrigerant precharge (oz/kg)		55.38 / 1.57
Length for precharge (ft/m)		24.6 / 7.5
Liquid side (in/mm)		1/4" / Ø6.35
Gas side (in/mm)		1/2" / Ø12.7
Additional Charge of Refrigerant (oz/ft)/(g/m)		0.16 / 15
Max. refrigerant pipe length (ft/m)		98.40 / 30
Max. difference in height (ft/m)		65.60 / 20

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Operation Range

ODU	Cooling (°F / °C)	-22 ~ 122 / -30 ~ 50
	Heating (°F / °C)	-22 ~ 75 / -30 ~ 24
IDU	Cooling (°F / °C)	60 ~ 90 / 16 ~ 32
	Heating (°F / °C)	32 ~ 86 / 0 ~ 30

System Data

Refrigerant type		R454B
Design pressure (PSIG)		550 / 340
Dehumidification (pts/hr)		N/A
IDU	Fan motor model	ZKFN-165-10-1L
	Fan motor qty (ea)	1
	Fan motor input (W)	185.0
	Fan speed, H / M / L (RPM)	1100 / 900 / 700
	Air flow, Turbo / H / M / L / Si (CFM)	600.4 / 559.2 / 500.3 / 441.4 / N/A
	Noise level, Turbo / H / M / L / Si (dB(A))	N/A / 40 / 39 / 38 / 32
	Net weight (lbs / kg)	54.45 / 24.7
	Gross weight (lbs / kg)	66.14 / 30
ODU	ESP range (Pa)	0-100
	Drain pipe outside diameter in. (mm)	OD 0.98 (25)
	Fan motor model	ZKFN-80-8-3
	Fan motor qty (ea)	1
	Fan motor input (W)	115
	Fan motor speed, H / / L (RPM)	780 / 700 / 550
	Air flow, Max (CFM)	1765.8
	Noise level, Max (dB(A))	59.0
Accessories	Net weight (lbs / kg)	99.87 / 45.3
	Gross weight (lbs / kg)	107.80 / 48.9

Accessories

BMS-WT2-XXD (8733962707)	Wired Wall Thermostat
BMS-DRY-XXD (8733969263)	24V Interface
BMS-PK2-7X1 (8733951011)	25ft Lineset

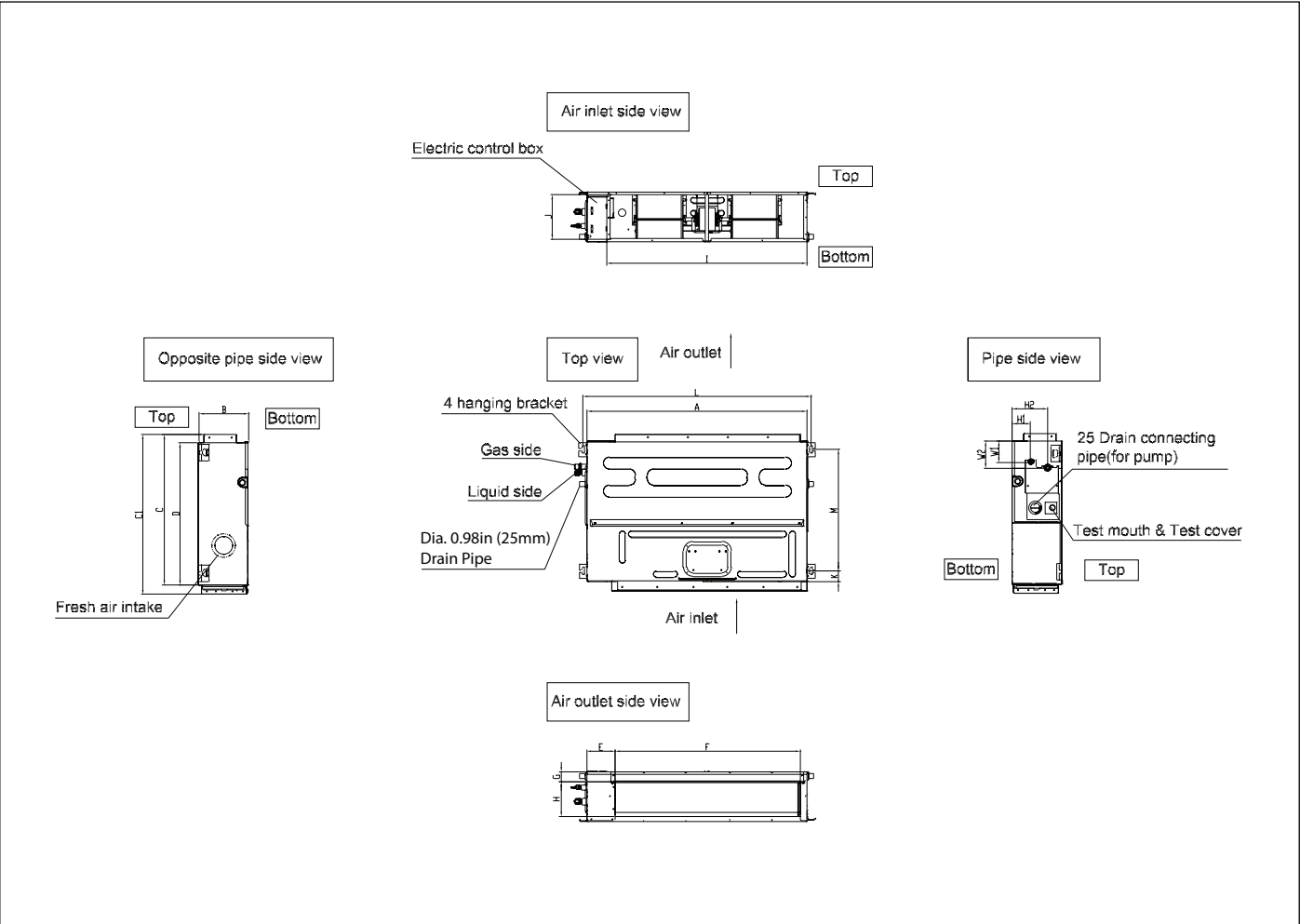
Bosch Thermotechnology Corp.
 Londonderry, NH • Watertown, MA • Ft. Lauderdale, FL

Tel: 1-800-283-3787 www.bosch-homecomfort.us

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18K Btu Single Zone Max Performance System - Ducted IDU



Indoor Unit Dimensions and Connections

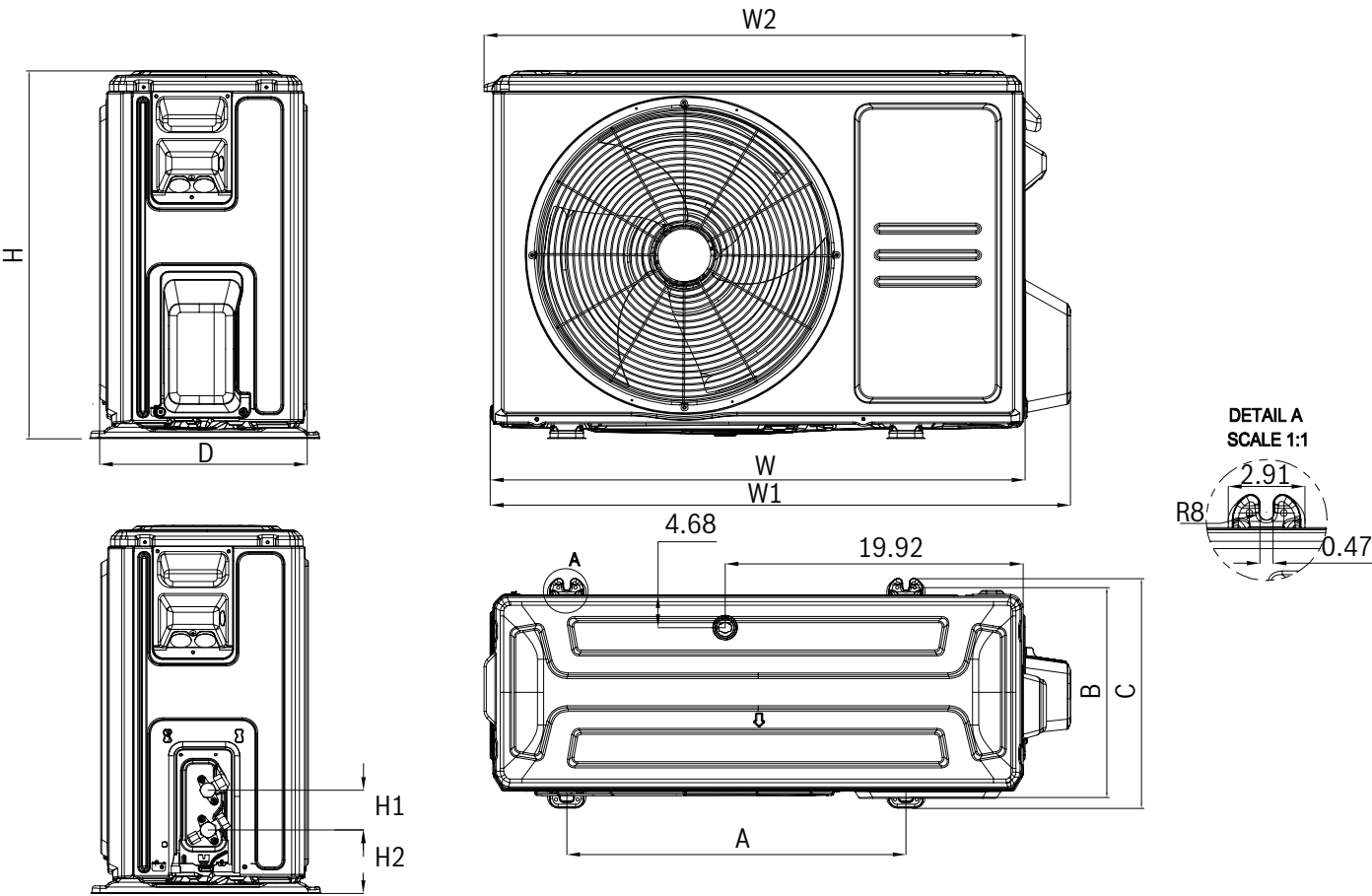


Model	Outline dimension (in)				Air outlet dimension (in)				Return air (in)			Mounting (in)		Gas and liquid valve (in)			
	A	B	C	D	E	F	G	H	I	J	K	L	M	H1	H2	W1	W2
BMS500-AAU018-1AHDXD	34.7	8.3	26.5	23.6	5.5	27.8	2.0	5.4	30.8	7.5	1.6	36.2	20.0	3.1	5.8	3.5	4.4

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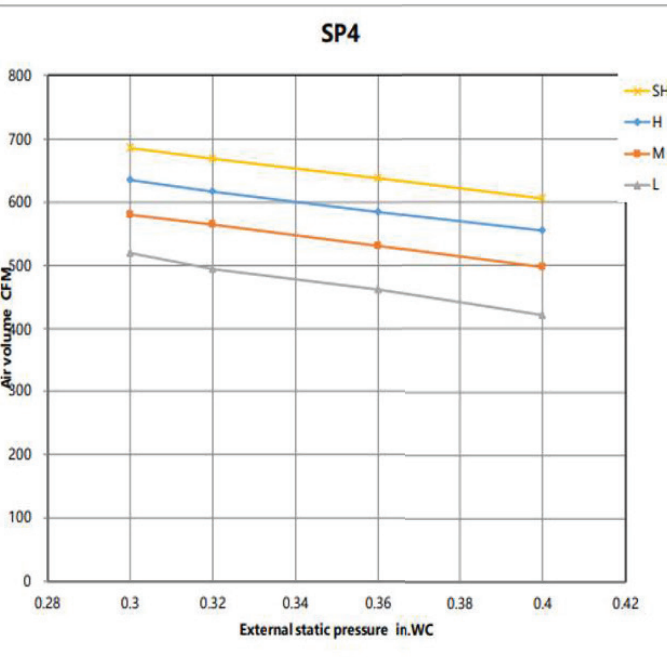
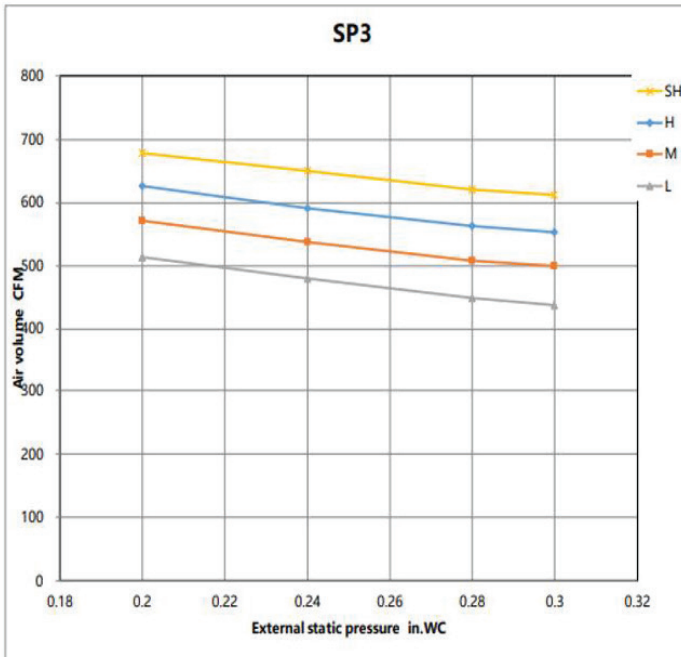
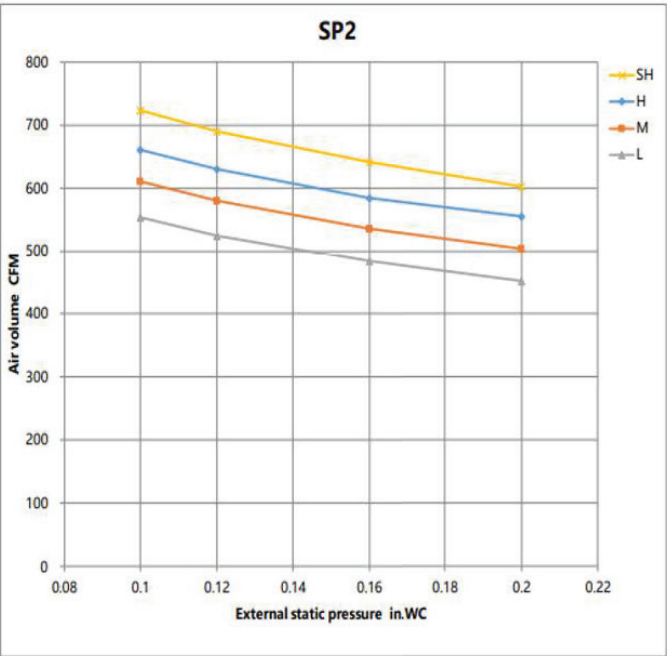
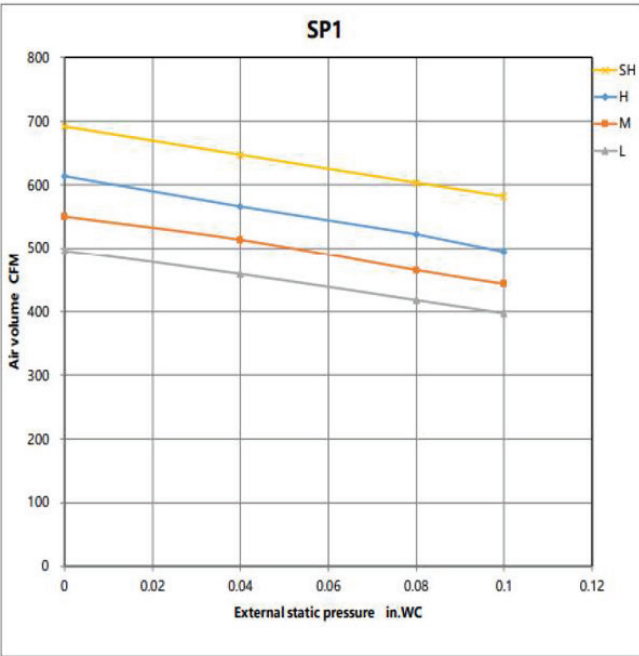


Outdoor Unit Dimensions and Connections

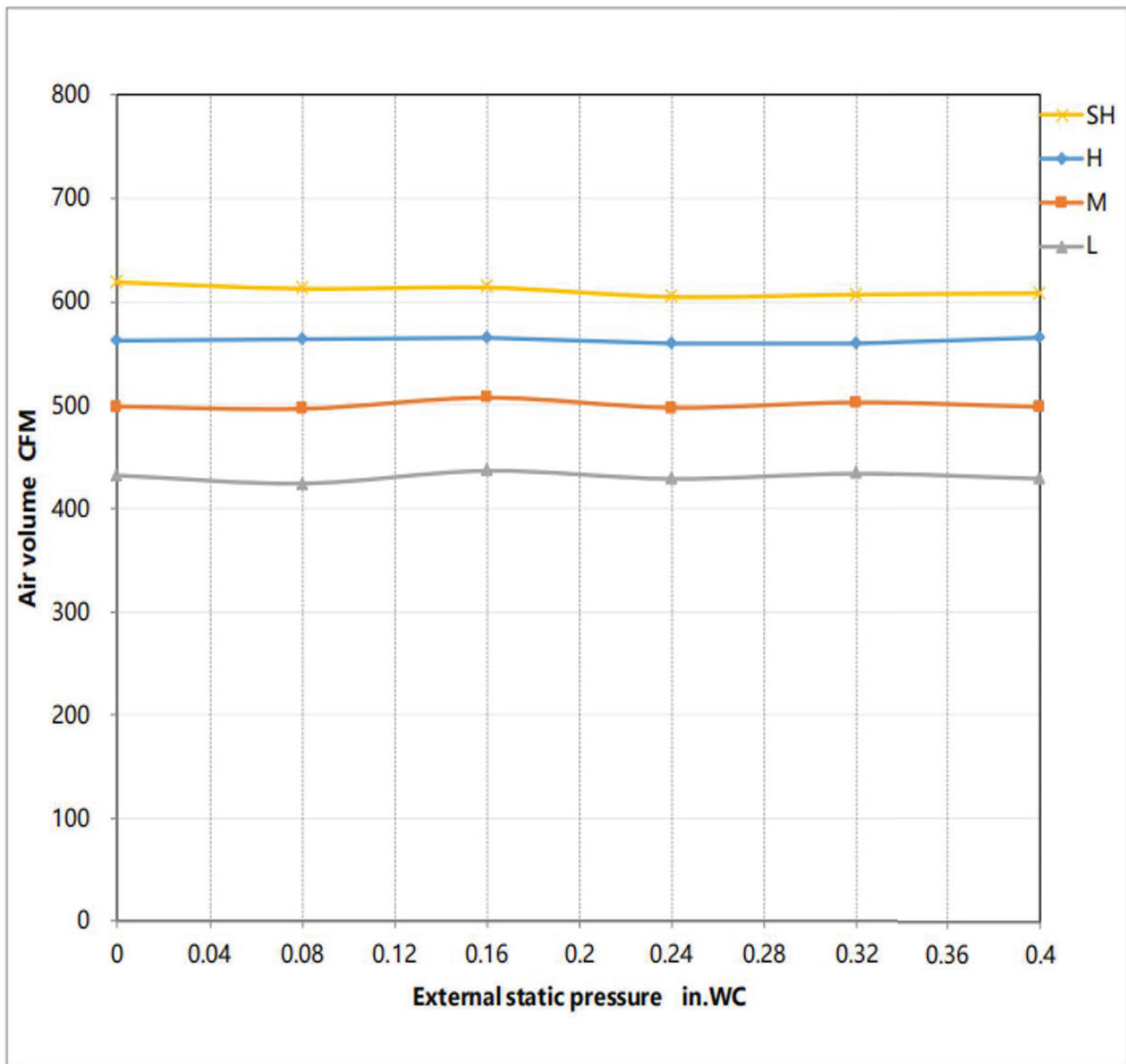


Model	W (in)	D (in)	H (in)	W1 (in)	W2 (in)	H1 (in)	H2 (in)	A (in)	B (in)	C (in)
BMS500-AAS018-1CSXHD	35.04	13.46	26.50	37.6	35.24	2.36	4.25	26.1	13.7	14.96

Air Volume



Air Volume



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Cooling Performance - Indoor Dry Bulb Temperature at 600 CFM																					
ODU DB(°F)	IDU WB (°F)	60.8°F(16.0°C)					64.4°F(18.0°C)					67°F(19.4°C)					71.6°F(22.0°C)				
	IDU DB (°F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)
-22°F (-30.0°C)	TC	14040	14034	14234	14293	14686	14649	14637	14628	14626	14614	15087	15068	15054	15052	15037	15914	15910	15877	15871	15837
	S/T	0.80	0.91	0.99	0.99	0.99	0.64	0.75	0.84	0.86	0.97	0.55	0.65	0.74	0.75	0.86	0.39	0.47	0.55	0.57	0.67
	PI	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76
-13°F (-25.0°C)	TC	15210	15203	15420	15484	15910	15870	15857	15847	15845	15832	16344	16324	16308	16306	16290	17240	17235	17200	17193	17157
	S/T	0.80	0.91	1.00	1.00	1.00	0.65	0.75	0.85	0.86	0.97	0.55	0.65	0.74	0.76	0.86	0.39	0.47	0.56	0.57	0.67
	PI	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.78	0.78	0.78	0.78	0.79	0.79	0.79	0.79	0.79	0.79
-4°F (-20.0°C)	TC	16380	16373	16606	16675	17134	17090	17076	17066	17064	17050	17601	17579	17563	17560	17543	18566	18561	18523	18516	18477
	S/T	0.80	0.92	1.00	1.00	1.00	0.65	0.76	0.85	0.87	0.98	0.55	0.65	0.74	0.76	0.86	0.39	0.48	0.56	0.57	0.67
	PI	0.82	0.82	0.82	0.82	0.81	0.81	0.81	0.81	0.81	0.81	0.81	0.81	0.81	0.81	0.81	0.82	0.82	0.82	0.82	0.82
0°F (-17.8°C)	TC	16924	16914	17228	17300	17775	17650	17630	17617	17615	17792	18169	18143	18120	18118	18109	19146	19141	19140	19142	19155
	S/T	0.81	0.93	1.00	1.00	1.00	0.66	0.77	0.86	0.88	1.00	0.55	0.66	0.75	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.82	0.83	0.82	0.82	0.82	0.82	0.82	0.82	0.82	0.82	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.83
5°F (-15.0°C)	TC	17409	17469	17868	17951	18504	18226	18230	18231	18250	18522	18808	18813	18817	18818	18843	19912	19906	19916	19918	19927
	S/T	0.82	0.94	1.00	1.00	1.00	0.66	0.78	0.87	0.88	1.00	0.56	0.67	0.76	0.78	0.88	0.39	0.48	0.57	0.59	0.69
	PI	0.83	0.83	0.83	0.83	0.84	0.83	0.84	0.84	0.84	0.84	0.84	0.84	0.84	0.84	0.84	0.86	0.86	0.86	0.86	0.86
14°F (-10.0°C)	TC	18537	18528	18906	18995	19584	19458	19441	19427	19425	19607	20124	20099	20079	20076	20055	21399	21392	21351	21344	21298
	S/T	0.81	0.93	1.00	1.00	1.00	0.65	0.76	0.86	0.87	1.00	0.55	0.66	0.75	0.76	0.87	0.39	0.48	0.56	0.57	0.67
	PI	0.78	0.79	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.79	0.79	0.79
17°F (-8.3°C)	TC	18868	18930	19367	19462	20094	19807	19816	19819	19836	20115	20483	20491	20499	20500	20526	21785	21779	21775	21779	21798
	S/T	0.82	0.93	1.00	1.00	1.00	0.66	0.77	0.87	0.88	1.00	0.55	0.66	0.76	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.77	0.77	0.77	0.77	0.77
23°F (-5.0°C)	TC	18777	18858	19346	19441	20082	19728	19734	19736	19762	20104	20409	20416	20421	20421	20454	21700	21694	21709	21712	21725
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78	0.89	0.39	0.48	0.57	0.59	0.69
	PI	0.78	0.78	0.78	0.78	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.81	0.81	0.81	0.81	0.81
35°F (1.7°C)	TC	18772	18772	18883	18935	19546	19649	19648	19650	19650	19702	20264	20263	20264	20264	20266	21405	21400	21399	21398	21400
	S/T	0.78	0.88	0.97	0.99	1.00	0.63	0.73	0.82	0.83	0.93	0.54	0.63	0.72	0.73	0.83	0.39	0.46	0.54	0.56	0.65
	PI	0.77	0.77	0.78	0.78	0.81	0.82	0.82	0.82	0.82	0.82	0.85	0.85	0.85	0.85	0.85	0.93	0.93	0.93	0.93	0.93
41°F (5.0°C)	TC	18580	18580	18728	18858	19416	19444	19444	19445	19445	19513	20051	20049	20051	20051	20052	21176	21171	21169	21168	21171
	S/T	0.78	0.89	0.98	1.00	1.00	0.64	0.74	0.83	0.84	0.94	0.54	0.64	0.72	0.74	0.84	0.39	0.47	0.55	0.56	0.66
	PI	0.80	0.80	0.81	0.82	0.85	0.85	0.85	0.85	0.85	0.86	0.89	0.89	0.89	0.89	0.89	0.97	0.97	0.97	0.97	0.97
47°F (8.3°C)	TC	18388	18400	18647	18730	19284	19240	19240	19240	19240	19329	19838	19836	19837	19837	19837	20948	20943	20941	20940	20943
	S/T	0.79	0.90	1.00	1.00	1.00	0.64	0.75	0.84	0.85	0.96	0.54	0.65	0.73	0.75	0.85	0.39	0.47	0.55	0.57	0.67
	PI	0.83	0.84	0.85	0.86	0.89	0.89	0.89	0.89	0.89	0.90	0.93	0.93	0.93	0.93	0.93	1.02	1.01	1.01	1.01	1.01
50°F (10.0°C)	TC	18081	18579	19206	19317	20066	19102	19112	19245	19305	20085	19847	19873	19883	19885	20076	21215	21271	21326	21332	21359
	S/T	0.90	1.00	1.00	1.00	1.00	0.71	0.85	0.96	0.99	1.00	0.59	0.72	0.83	0.85	0.99	0.40	0.50	0.60	0.62	0.74
	PI	0.96	0.96	0.96	0.96	0.95	0.96	0.96	0.96	0.96	0.95	0.95	0.95	0.95	0.95	0.95	0.95	0.95	0.95	0.95	0.95
59°F (15.0°C)	TC	18149	18785	19452	19570	20367	19202	19228	19446	19591	20387	19994	20014	20021	20050	20404	21470	21513	21556	21560	21578
	S/T	0.92	1.00	1.00	1.00	1.00	0.73	0.86	1.00	1.00	1.00	0.60	0.73	0.85	0.86	1.00	0.40	0.51	0.61	0.63	0.76
	PI	1.02	1.02	1.02	1.02	1.03	1.02	1.02	1.02	1.02	1.03	1.03	1.03	1.03	1.03	1.03	1.03	1.03	1.03	1.03	1.03
68°F (20.0°C)	TC	18207	18984	19688	19813	20654	19275	19318	19709	19834	20674	20108	20123	20159	20177	20691	21666	21707	21741	21744	21755
	S/T	0.95	1.00	1.00	1.00	1.00	0.74	0.89	1.00	1.00	1.00	0.61	0.75	0.87	0.89	1.00	0.40	0.51	0.63	0.65	0.78
	PI	1.08	1.09	1.09	1.09	1.10	1.09	1.09	1.09	1.09	1.10	1.10	1.10	1.10	1.10	1.10	1.11	1.11	1.11	1.11	1.11
77°F (25.0°C)	TC	18249	19128	19877	20009	20903	19347	19408	19898	20031	20924	20223	20245	20297	20320	20942	21841	21908	21958	21964	21984
	S/T	0.97	1.00	1.00	1.00	1.00	0.75	0.90	1.00	1.00	1.00	0.61	0.76	0.88	0.90	1.00	0.40	0.52	0.63	0.65	0.79
	PI	1.15	1.16	1.16	1.16	1.17	1.16	1.16	1.16	1.16	1.17	1.17	1.17	1.17	1.17	1.17	1.18	1.18	1.18	1.18	1.18
82°F (27.8°C)	TC	18053	18974	19740	19873	20783	19130	19204	19757	19896	20805	20020	20045	20108	20132	20824	21646	21727	21784	21790	21813
	S/T	0.99	1.00	1.00	1.00	1.00	0.76	0.91	1.00	1.00	1.00	0.62	0.77	0.89	0.92	1.00	0.40	0.52	0.64	0.66	0.80
	PI	1.25	1.25	1.26	1.26	1.26	1.25	1.25	1.26	1.26	1.26	1.26	1.26	1.26	1.26	1.27	1.28	1.27	1.27	1.27	1.27
86°F (30.0°C)	TC	17678	18560	19319	19454	20360	18643	18729	19341	19476	20381	19526	19549	19620	19649	20399	21139	21221	21271	21277	21296
	S/T	1.00	1.00	1.00	1.00	1.00	0.77	0.93	1.00	1.00	1.00	0.62	0.78	0.91	0.93	1.00	0.40	0.52	0.64	0.66	0.81
	PI	1.39	1.40	1.40	1.40	1.41	1.40	1.40	1.40	1.40	1.41	1.41	1.41	1.41	1.41	1.41	1.43	1.42	1.42	1.42	1.42
95°F (35.0°C)	TC	16698	16789	17331	17434	18123	17747	177													

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Heating Performance - Indoor Dry Bulb Temperature at 600 CFM										
Outdoor DB(°F)	TC Total Heating Capacity(BTU/hr)					PI Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-22°F(-30.0°C)	10532	10322	10270	10229	10137	1.91	1.93	1.93	1.93	1.94
-13°F(-25.0°C)	13108	12846	12781	12730	12616	2.07	2.08	2.09	2.09	2.10
-4°F(-20.0°C)	16323	15997	15917	15853	15711	2.25	2.27	2.27	2.28	2.29
0°F(-17.8°C)	17286	16940	16856	16788	16637	2.37	2.39	2.39	2.40	2.41
5°F(-15.0°C)	18489	18120	18029	17957	17795	2.52	2.54	2.54	2.55	2.56
14°F(-10.0°C)	20116	19714	19616	19537	19361	2.49	2.51	2.51	2.52	2.53
17°F(-8.3°C)	20659	20246	20144	20064	19883	2.48	2.50	2.50	2.51	2.52
22°F(-5.6°C)	20750	20335	20234	20153	19971	2.36	2.38	2.39	2.39	2.40
27°F(-2.8°C)	20842	20425	20323	20241	20059	2.25	2.27	2.27	2.27	2.28
32°F(0.0°C)	20933	20514	20412	20330	20147	2.13	2.15	2.15	2.16	2.17
35°F(1.7°C)	20988	20568	20465	20383	20200	2.06	2.08	2.09	2.09	2.10
37°F(2.8°C)	21133	20710	20607	20524	20340	2.02	2.03	2.04	2.04	2.05
42°F(5.6°C)	21496	21066	20961	20877	20689	1.90	1.91	1.92	1.92	1.93
44.6°F(7.0°C)	21641	21208	21102	21018	20829	1.85	1.86	1.87	1.87	1.88
47°F(8.3°C)	21859	21422	21315	21230	21038	1.78	1.79	1.80	1.80	1.81
52°F(11.1°C)	20966	20547	20444	20362	20179	1.58	1.59	1.60	1.60	1.60
57°F(13.9°C)	20073	19672	19573	19495	19320	1.38	1.39	1.39	1.40	1.40
59°F(15.0°C)	19716	19322	19225	19148	18976	1.30	1.31	1.31	1.32	1.32
62°F(16.7°C)	19180	18797	18703	18628	18460	1.18	1.19	1.19	1.20	1.20
64.4°F(18.0°C)	18752	18377	18285	18212	18048	1.09	1.09	1.10	1.10	1.10
69.8°F(21.0°C)	16610	16278	16197	16132	15987	0.87	0.88	0.88	0.88	0.89
75.2°F(24.0°C)	13395	13127	13061	13009	12892	0.66	0.66	0.66	0.66	0.67

Bolded values were tested. All other values are simulated and for reference only.